An independent double-difference tomography implementation: synthetic reconstruction and limits of manuscript reproduction

Independent computational reproduction

October 2026

Abstract

Double-difference tomography uses paired seismic arrivals to reduce shared travel-time errors while estimating source locations and velocity jointly. Here we implement this inverse problem independently in C++ and test it against the written model of Zhang and Thurber (2003). The code combines variational ray tracing, trilinear velocity interpolation, joint linearization, and damped, smoothed least squares. We reconstruct the specified synthetic velocity profile and generate new event and station geometries. Consequently, the true-model panel is independently reproduced, whereas synthetic inversion panels are partial reproductions. The Hayward field-data figures depend on observations excluded by the initial project instructions and are not reproduced. All numerical figures in this report use saved output from this implementation. The result is a reproducible method investigation, not a recovery of the published Hayward velocity model. The resumed work permits authoritative observational inputs. A later section records successful archive access and ray-kernel corrections; the reported inverse solutions remain the baseline results.

1 Scientific scope and evidence

The target paper is Zhang and Thurber (2003), *Bulletin of the Seismological Society of America* 93, 1875–1889, doi:10.1785/0120020190. Its written equations, experiment descriptions, and captions are the scientific specification. The publisher page returned HTTP 403; the complete manuscript transcription posted by an author on ResearchGate provided the text. The baseline calculations used no original model code, source observations, synthetic arrival files, plotting scripts, or figure files. No published plot was digitized. No published image appears in this report.

The initial exclusion of source observations had a direct scientific consequence. A physical forward model can generate new arrival times for a specified geometry, but it cannot recover the historical earthquake realization and picking errors from a prose description. We therefore separate a reconstructible velocity specification, an independent synthetic inverse experiment, and unavailable field-data inversions. The panel audit preserves all eleven manuscript figures rather than replacing the missing field results with invented observations. No supplement is listed for this article on the original SSA issue contents or its original abstract page. No supplementary figures were identified. The official issue contents marks supplements for other articles explicitly; `reference/specification_notes.md` records those checks.

2 Forward problem and joint inversion

Assume an isotropic, stationary medium and high-frequency geometric P rays. Coordinates are in km, velocity in km/s, and time in s; depth is positive downward. The predicted arrival of event i at station k is

$$T_{ik} = \tau_i + \int_{\Gamma_{ik}} \frac{1}{v(\mathbf{x})}\, ds. \tag{1}$$

Here τ_i is origin time and Γ_{ik} is a stationary travel-time path with fixed endpoints. The calculation includes no reflected or converted phases, attenuation, or wave-amplitude effects. Paths remain inside the rectangular model domain. There are no periodic or reflecting boundary conditions.

Nodal velocities v_n are interpolated trilinearly. Write the update as $q_n = \delta \log(1/v_n)$, so that the nonlinear velocity update is $v_n^{\text{new}} = v_n \exp(-q_n)$. The absolute residual linearization is

$$r_{ik} \simeq \mathbf{g}_{ik} \cdot \delta\mathbf{x}_i + \delta\tau_i + \sum_n K_{ikn} q_n, \qquad K_{ikn} = v_n \int_{\Gamma_{ik}} \frac{w_n(\mathbf{x})}{v(\mathbf{x})^2}\, ds. \tag{2}$$

w_n is the interpolation weight. The source derivative $\mathbf{g}_{ik}$ is evaluated from the differentiated endpoint segment integral. Absolute arrivals constrain structure outside clusters and the common source shift.

For a measured event-pair difference d_{ijk} at a common station, use

$$r_{ijk}^{\text{DD}} = d_{ijk} - (T_{ik} - T_{jk}), \qquad A_{ijk}^{\text{DD}} = A_{ik} - A_{jk}. \tag{3}$$

Velocity sensitivities from both events enter the difference, subject to the 5 m sparsification rule described below. This includes changes in path anomalies with source location. Shared station timing errors cancel in differenced observations, whereas independent pick errors do not. Catalog differences share errors with each other and with absolute picks; our weighting follows the staged formulation without whitening this covariance.

The linear step minimizes a weighted residual norm with first-difference regularization of the final log slowness and damping of the column-scaled increment. A first-difference row divides the difference between adjacent nodes by their separation. Columns are normalized to unit Euclidean norm before damping rows are added. An independently written LSQR implementation solves the sparse system without forming normal equations. Absolute, catalog-difference, and precise-difference rows receive stage weights $(1, 0.1, 0.01)$, $(0.1, 1, 0.01)$, and $(0.001, 0.01, 1)$, respectively. Smoothing and damping choices are explicit in the configuration. After a rejected update, identical repetitions within the same stage are skipped, while every later weight stage is retained.

Each accepted nonlinear step must decrease the current weighted misfit plus roughness. Weights are frozen within that line search. Trial source locations are projected onto the grid domain; velocities remain positive by logarithmic parameterization. No published solution is used to choose an update. Sources at boundaries and failed updates are counted in the results because they can indicate model error or weak constraints.

2.1 Ray integration and numerical substitutions

The baseline ray tracer minimizes discretized Fermat time with limited-memory BFGS and Armijo backtracking, starting from a straight path. Each segment is split at grid-plane crossings and integrated with two-point Gaussian quadrature. This independent variational method replaces the paper's pseudo-bending implementation. It searches for a local stationary path, with no guarantee of finding every competing first-arrival branch. Search directions and convergence gradients respect active model boundaries. Near-coincident interior nodes are merged at a numerical tolerance far below ray spacing, with the quasi-Newton history reset after such topology changes. Independent rays are traced in parallel with OpenMP.

The reference configuration uses 1 km initial ray spacing and 180 ray iterations, 18 nonlinear iterations in three equal stages, smoothing weight 5, and scaled damping 0.1. These choices are not all specified by the paper. Distance and residual biweights, a 5 m differential path-coefficient cutoff, regularization scaling, line search, and the residual-scale estimator are fully defined in `METHODS.md` and the source. A damping or smoothing number cannot be transferred between implementations without its scaling convention. Our equal-direction gradient penalty should therefore not be assumed identical to the original regularization.

3 Independent synthetic experiment

The true medium is constant with depth and along strike. Its cross-strike profile is 6 km/s for $x \leq 0$, falls linearly to 4 km/s at $x = 1$ km, remains 4 km/s through $x = 5$ km, rises linearly to 5 km/s at $x = 6$ km, and increases linearly to 6 km/s at $x = 38$ km. The paper describes sharp gradients over $x = 0$–1 and 5–6 km; linear interpolation across these two transitions is our explicit assumption. Figure 1 is generated from saved model sampling and reconstructs the written velocity specification with this assumption, without using published pixel values.

The inversion grid follows the coordinates in manuscript caption 2: $x = (-35, -15, 0, 2, 4, 6, 20, 35)$ km, $y = (-60, -40, -20, 0, 20, 40)$ km, and $z = (0, 3, 7, 11, 16)$ km. The forward grid adds the profile breakpoints $x = 1$ and 5 km. This prevents the data generator from using exactly the same coarse representation as the inverse problem.

We generate 120 sources and 32 surface stations independently with seed 2003. Half the sources sample a 12 km wide region around the trough, and half sample a broader 40 km interval; all sample the model volume in depth and along strike. Stations use a jittered surface array. Figure 2 shows the complete generated geometry. These are not the paper's Loma Prieta observations. The starting velocity is homogeneous at 5.5 km/s, substituting for the paper's referenced layered starting model, whose values are not tabulated in the target article. Independent starting source-coordinate errors have standard deviation 0.6 km, with an initial depth restriction of 0.5–15.5 km. Starting origin-time errors have standard deviation 0.05 s.

We round forward travel times to 0.01 s, add independent 0.04 s Gaussian pick noise and a station-constant uniform bias between −0.3 and 0.3 s. Catalog differences subtract these noisy picks. Precise synthetic differences subtract rounded forward times before added pick noise. This explicitly resolves an ambiguity in the text's noise ordering in favor of precise pair differences. Every event is observed at every station, and pairs within 20 km are selected. The saved arrival table separates exact time, rounding, station bias, pick noise, and final observed time. Subtracting two separately rounded arrivals permits differential rounding errors approaching 0.01 s. The paper quotes 0.005 s for its differential data; this discrepancy remains a limitation of our explicit rounding order.

3.1 Starting-model sensitivity

The target paper does not identify a side of the Dietz–Ellsworth model. USGS Open-File Report 93-578, Table 4, tabulates two gradient adaptations attributed to that work. We sample each named profile (LOM and LON) onto the same inversion-depth grid and repeat DD inversion using otherwise identical inputs. The adaptation footnote prevents identifying either as the confirmed target start. This comparison measures sensitivity to two written alternatives without inventing their values.

Start	Median 3D error	Mean $\lvert\Delta v\rvert$	Interior $\lvert\Delta v\rvert$	R_{abs}	R_{prec}
Homogeneous	0.8222	0.3050	0.2135	0.3526	0.4390
LOM	1.4316	0.4036	0.2589	0.3752	0.4370
LON	1.1469	0.3606	0.2444	0.3780	0.4348

Table 1: Starting-model sensitivity of the independent DD experiment. LOM and LON are written gradient adaptations in USGS OFR 93-578. All runs share geometry, true velocity, and observation noise. Median three-dimensional location errors are in km, mean absolute velocity errors are in km/s, and arrival residual RMS values are in s. The first velocity column uses all saved slices; the interior column restricts x to $[-15, 20]$ km, y to $[-40, 20]$ km, and depths to 3–11 km.

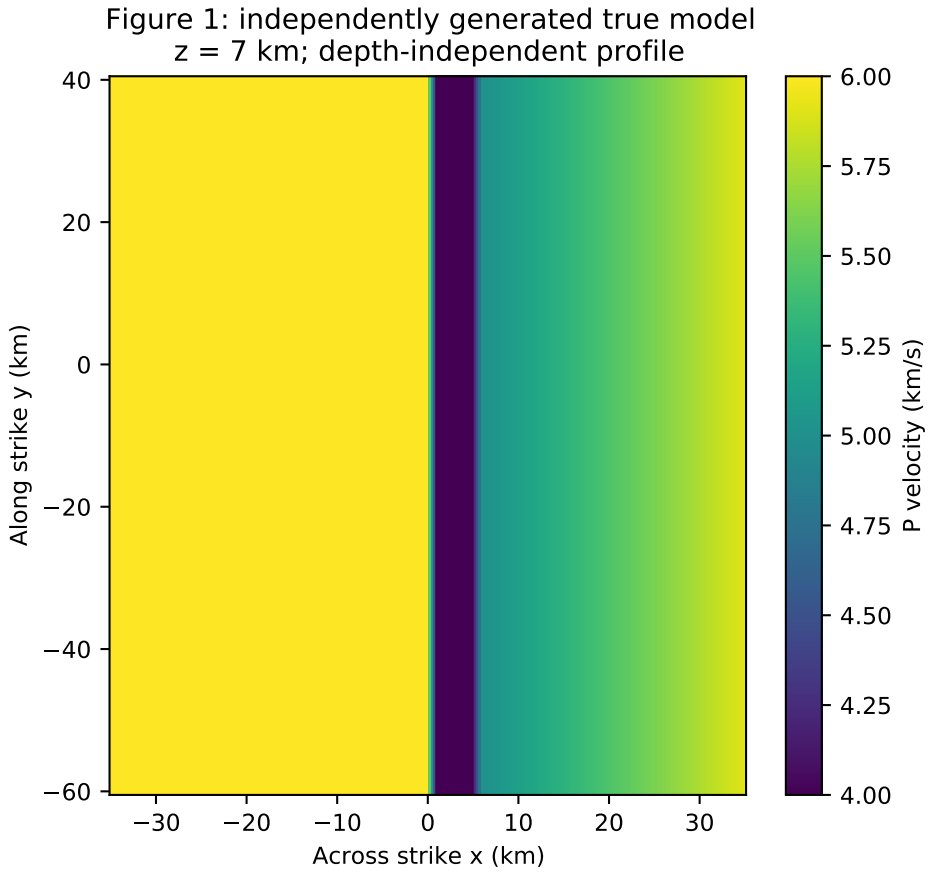


Figure 1: Manuscript Figure 1 reconstruction: true P velocity from the written sandwich profile, sampled at 7 km depth. The model is independent of depth; colors show velocity in km/s. All plotted samples are saved C++ output. The trough occupies $x = 1$–5 km and has velocity 4 km/s.

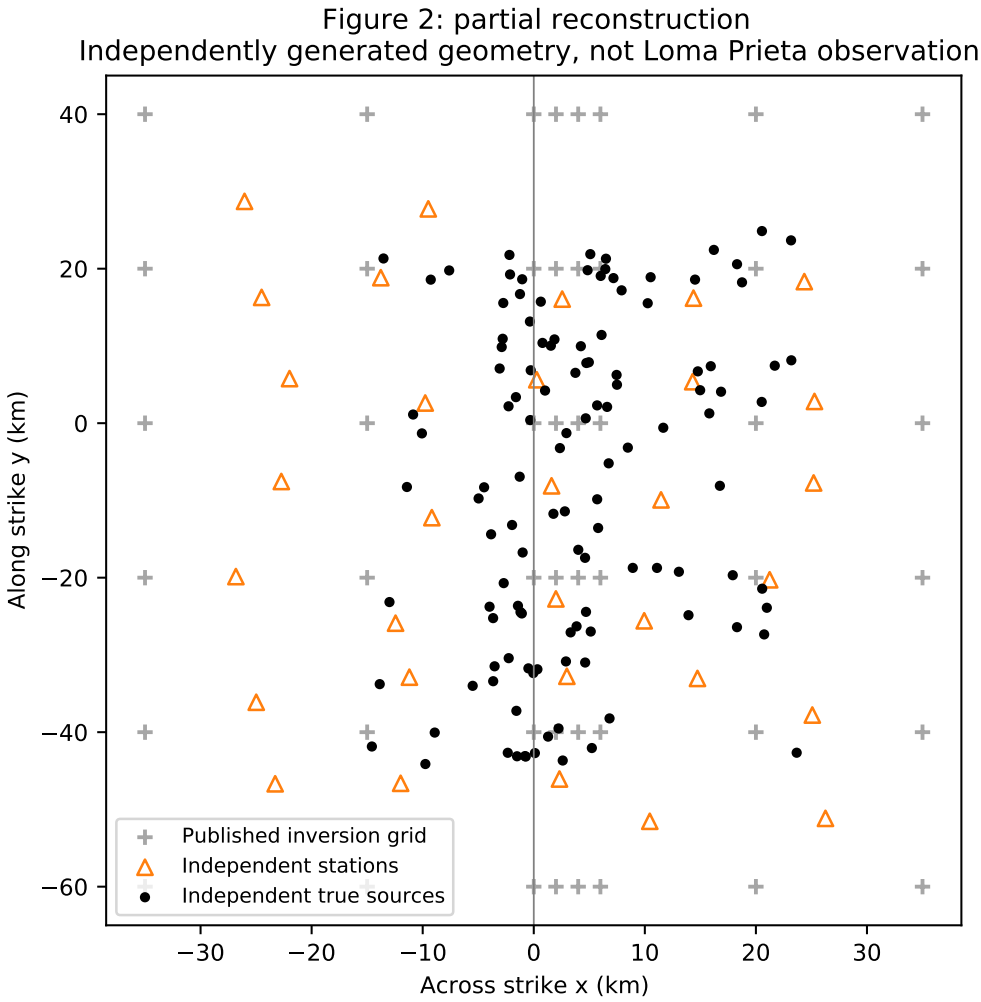


Figure 2: Partial reconstruction of manuscript Figure 2. Crosses are the written inversion grid, triangles are independent stations, and dots are independent true sources. The coordinates are experimental design choices, not estimates extracted from the published map.

4 Validation and numerical results

All 32 implementation consistency checks passed. Across 32 deterministic source–receiver pairs, halving the ray segment spacing and increasing the iteration limit changed travel times by a median of 0.054 ms and a maximum of 0.892 ms. This check does not establish global ray optimality.

In the small known-model recovery check, the median three-dimensional location error decreased from 0.3035 to 0.0343 km; the final absolute-time RMS was 0.0034 s. Only 0.01 s time quantization remains in this check.

Before the main inversions, the median three-dimensional source error was 1.0285 km. The homogeneous 5.50 km/s initial model had mean absolute velocity errors of 0.456 km/s over the full domain and 0.544 km/s in the interior, evaluated at the same saved slice samples as the recovered models.

The standard run ended with median three-dimensional source error 1.7069 km, 0 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 2.41 to 7.05 km/s. LSQR converged in 10 of 10 solves, with worst relative normal residual 8.35e-09. The dd run ended with median three-dimensional source error 0.8222 km, 0 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 3.74 to 6.41 km/s. LSQR converged in 18 of 18 solves, with worst relative normal residual 8.5e-09. The loc1d run ended with median three-dimensional source error 2.9856 km, 11 sources on a model boundary, 1 zero-update iterations, and nodal velocities ranging from 5.50 to 5.50 km/s. LSQR converged in 15 of 15 solves, with worst relative normal residual 9.64e-09. The loc3d run ended with median three-dimensional source error 0.1974 km, 14 sources on a model boundary, 0 zero-update iterations, and nodal velocities ranging from 4.00 to 6.00 km/s. LSQR converged in 18 of 18 solves, with worst relative normal residual 9.32e-09.

Relative to standard tomography, DD changed the full-domain mean absolute velocity error by -0.269 km/s and the median depth error by -0.996 km. Negative differences indicate improvement. These comparisons refer only to the independent experiment. The analytic homogeneous joint-inversion recovery gates **PASSED**. Median three-dimensional location error changed from 0.1821 to 0.0210 km, while median velocity error at the 203 illuminated nodes changed from 0.3000 to 0.0080 km/s. The absolute residual RMS changed from 0.35209 to 0.00635 s. The precise residual RMS changed from 0.09192 to 0.00031 s. There were 12 accepted velocity updates, 12 accepted location updates, and 0 sources on a boundary. LSQR met its normal-residual tolerance in 12 of 12 solves; the worst relative normal residual was 9.15e-09.

The fixed recovery criteria require positive finite velocities, no boundary sources, nonincreasing accepted frozen-weight objectives, at least tenfold reductions of both absolute and precise residual RMS, lower median location and illuminated-node velocity errors, and nonzero updates to both locations and velocity. Illumination means accumulated interpolated path length greater than 10^{-9} km on final rays; it does not establish resolution or recovery at every node. The arrivals are analytic and unrounded. This test examines a coupled homogeneous limiting case; it does not establish heterogeneous model recovery. Ray diagnostics compare production traces with half-spacing, 1000-iteration traces at the saved true, final standard, and final DD models and event locations. Source derivatives are compared with central differences of refined travel times using 0.01 km endpoint perturbations. A diagnostic exceedance uses the existing single-ray tolerance of 0.003 s/km; it is not a demonstrated inversion-error bound.

For the truth model, 32 rays had median and maximum absolute refinement changes of 0.054 and 0.892 ms. Production/refined convergence counts were 15/20 out of 32; maximum projected gradients were 0.00868/0.000668 s/km and maximum raw gradients were 0.00868/0.000668 s/km. Thus 17/12 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives, maximum production/refined errors were 0.000584/0.000599 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

For the standard model, 32 rays had median and maximum absolute refinement changes of 0.155 and 0.684 ms. Production/refined convergence counts were 1/3 out of 32; maximum projected gradi-

ents were 0.0118/0.00645 s/km and maximum raw gradients were 0.0118/0.00645 s/km. Thus 31/29 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives, maximum production/refined errors were 0.000746/0.000328 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

For the dd model, 32 rays had median and maximum absolute refinement changes of 0.100 and 0.861 ms. Production/refined convergence counts were 1/8 out of 32; maximum projected gradients were 0.0115/0.014 s/km and maximum raw gradients were 0.0115/0.014 s/km. Thus 31/24 production/refined traces failed the optimizer stopping criterion. Across 24 eligible source-coordinate derivatives, maximum production/refined errors were 0.000589/0.000179 s/km, with 0/0 components exceeding the diagnostic tolerance; 0 components were skipped at boundaries.

Raw and projected gradients differ at active domain boundaries. Small changes in travel time do not establish derivative convergence, and unconverged refined traces are not exact references. These deterministic samples cannot establish global first-arrival optimality or the accuracy of every production ray.

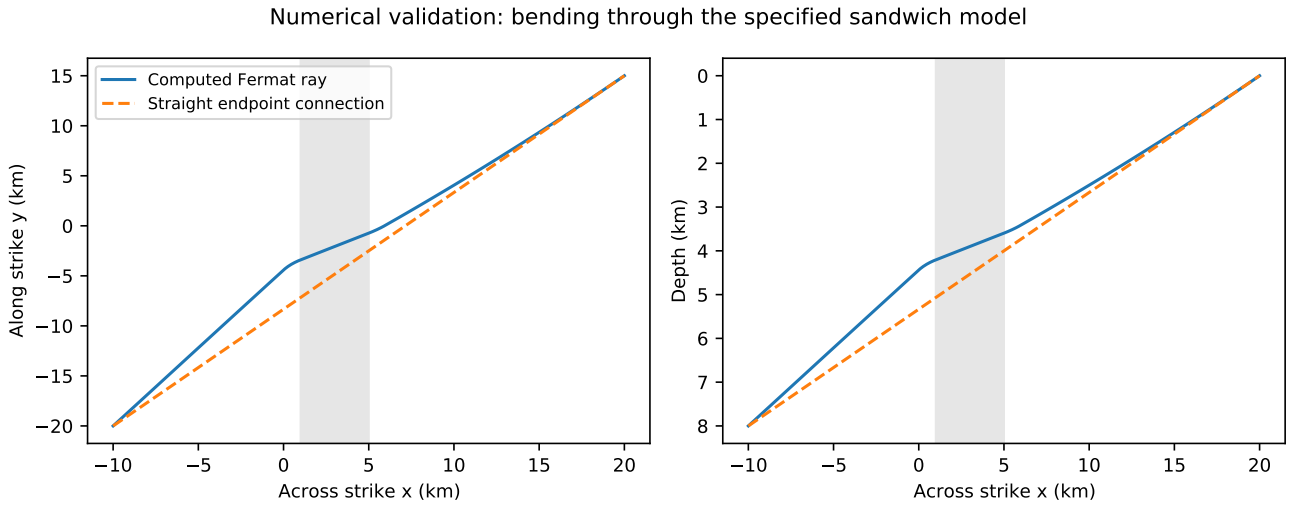


Figure 3: Saved numerical ray through the specified sandwich medium, compared with the straight connection between identical endpoints. The shaded region marks the low-velocity trough. Bending reduces calculated travel time. This is a validation figure, not a manuscript-panel reproduction.

The exact homogeneous travel-time solution, derivative checks, reciprocal ray times, station-error cancellation, and known least-squares solution test different parts of the implementation (Appendix A). They do not establish uniqueness of the nonlinear inverse problem. A larger, deterministic ray sample tests the production settings against half-spacing, 1000-iteration traces. The source finite-difference checks are reported for both the true and inferred media. Small travel-time differences alone do not establish stationarity or accurate source derivatives. Raw gradients can also be nonzero at active constraints, so projected gradients and iteration limits are recorded separately. The homogeneous joint test generates arrivals analytically as distance divided by 6 km/s, independently of the ray tracer, and enables both source and velocity unknowns.

The cross-strike-only medium also admits an independent Snell-law reference for the validation ray, which crosses the layers monotonically. If R is the endpoint separation in the transverse plane, its conserved transverse slowness p satisfies

$$R = \int_{x_a}^{x_b} \frac{p\, v(x)}{\sqrt{1 - p^2 v(x)^2}}\, dx, \qquad T = \int_{x_a}^{x_b} \frac{1}{v(x)\sqrt{1 - p^2 v(x)^2}}\, dx. \tag{4}$$

This scalar reference tests both travel time and the source slowness vector without optimizing path vertices. At the 600-iteration setting, a source-perturbation sweep found finite-difference errors below 1.53×10^{-4} s/km at 0.1, 0.03, and 0.01 km, while the 0.001 km difference amplified ray-discretization and optimization errors to 0.00350 s/km. More iterations did not ensure finite-difference convergence. The consistency check therefore uses 0.01 km, matching the larger ray diagnostic sample; the derivative tolerance remains 0.003 s/km. The saved sweep in `data/development/` records this limitation. It is not evidence of derivative convergence in every inferred three-dimensional medium.

Method	Median $\lvert\Delta x\rvert$	Median $\lvert\Delta y\rvert$	Median $\lvert\Delta z\rvert$	R_{abs}	R_{cat}	R_{prec}
standard	0.5232	0.4525	1.5176	0.3565	0.4621	0.4619
dd	0.3690	0.3266	0.5217	0.3526	0.4399	0.4390
loc1d	2.0050	0.5019	0.8753	0.6342	0.5095	0.5081
loc3d	0.0103	0.0221	0.1955	0.1771	0.0578	0.0154

Table 2: Independent synthetic results. Coordinate errors are in km and arrival residual RMS values are in s. Residual RMS values are unweighted, so they remain comparable when inversion weights change. Fixed-model relocations are denoted loc1d (homogeneous) and loc3d (true synthetic grid). These fixed-model comparisons retain absolute arrivals and do not copy all gauge conventions of hypoDD. The axes are Cartesian, not the latitude/longitude axes of published Table 1.

Method	Domain	Median	Mean	Standard deviation
Initial	Full domain	0.500	0.456	0.318
Initial	Interior	0.500	0.544	0.393
standard	Full domain	0.443	0.574	0.506
standard	Interior	0.258	0.313	0.245
standard	Published reference	0.164	0.245	0.249
dd	Full domain	0.252	0.305	0.262
dd	Interior	0.153	0.214	0.181
dd	Published reference	0.136	0.178	0.164

Table 3: Absolute velocity errors in km/s. Full-domain statistics use saved regular samples on all five depth slices. The interior restricts x to $[-15, 20]$, y to $[-40, 20]$, and depths to 3–11 km. Published values are reference scales only: the article does not specify the averaging mask, and its geometry and starting model differ.

DD reduces the full-domain mean velocity error from the initial 0.456 to 0.305 km/s; standard tomography ends at 0.574 km/s. The DD value remains larger than the published reference of 0.178 km/s, under the differing sampling and starting conditions described above. Median three-dimensional location error improves from 1.029 to 0.822 km with DD, although its median depth error, 0.522 km, is slightly worse than the initial 0.477 km. Thus improvement depends on both the quantity and the averaging domain.

The recovered DD western-block–trough contrasts are 1.577 and 1.663 km/s at 3 and 7 km depth, respectively, compared with the specified 2 km/s. DD preserves more contrast than standard tomography but still smooths the trough. Its unweighted precise-difference RMS remains 0.439 s, compared with 0.462 s for standard tomography and 0.0154 s with the true velocity fixed. The DD residual is far above the approximately 0.01 s upper bound on differential rounding error: the nearly exact synthetic differences are not fitted to their precision. Grid representation, regularization, source–velocity trade-offs, and robust downweighting are possible contributors; these experiments do not isolate their effects. Improved model-error averages therefore do not imply a close fit to every event pair. Among the three tested starting profiles, the homogeneous start gives the smallest final location and velocity

Model	Depth (km)	Trough	West block	West minus trough
standard	3	4.440	5.561	1.121
standard	7	4.592	5.948	1.356
dd	3	4.202	5.779	1.577
dd	7	4.238	5.901	1.663
Specified true model	3, 7	4.000	6.000	2.000

Table 4: Spatial recovery of the specified trough and western block. All velocities and contrasts are in km/s. Arithmetic means use saved samples with $y \in [-20, 20]$ km, $x \in [1, 5]$ km for the trough, and $x \in [-15, 0]$ km for the western block, including window endpoints. These independently defined windows quantify recovery of the known 2 km/s contrast; they are not published recovered-model averages.

errors; this is a sensitivity result for the independent experiment, not identification of the paper's start. The LOM and LON runs end with median three-dimensional errors of 1.432 and 1.147 km, respectively, both worse than the initial 1.029 km.

All sampled travel-time refinement changes are below 0.9 ms, smaller than the 10 ms rounding interval and 40 ms random-pick scale. None of the sampled source derivatives exceeds the 0.003 s/km diagnostic tolerance. However, most rays in the inferred models do not meet the stricter optimizer stopping criterion, including many refined traces. These checks show close agreement between the numerical settings on sampled paths. They do not certify absolute accuracy, all derivatives, or the first-arrival branch selected by every ray.

Each Figure 3 counterpart shows the five generated grid-depth slices at 0, 3, 7, 11, and 16 km. The published captions do not enumerate plotted depths, so this count is not a verified reconstruction of the published figure layout.

5 Discrepancies and reproduction limits

The largest structural difference from the target experiment is its sampling geometry. The synthetic coordinates in the original paper come from actual seismicity and station locations, not from a distribution completely specified in the text. Replacing that geometry changes the null space, ray coverage, and source–velocity trade-off. A homogeneous initial velocity also changes the nonlinear trajectory relative to the referenced layered model. We cannot attribute a mismatch with a published error statistic solely to the inversion algorithm under these changes.

Our forward grid represents the trough edges more finely than the inversion grid. This is a deliberate test of model representation error. The 0.3 s station biases are larger than the random pick errors, and the inverse problem has no explicit station-static parameters. It must accommodate these errors through source shifts, near-station velocity, or residual misfit. Robust weighting may suppress genuine model mismatch as if it were an outlier. A lower weighted objective therefore does not by itself establish a more accurate velocity model; velocity and location errors against the independent truth are reported separately.

During development, rejecting an entire trial update when one source left the domain caused the fixed-model inversion to stall. Projecting individual trial source locations onto the domain corrected that numerical obstruction. Independent review also found that boundary directions within the ray optimizer could block tangential motion, and that collapsed ray segments could corrupt endpoint derivatives. Both cases now have numerical regression checks. A further check found roundoff gradients in coordinate directions where velocity was exactly constant. Ray optimization and column scaling amplified these into spurious source-depth sensitivities. Evaluating each interpolation-gradient component with paired nodal differences preserves this symmetry exactly without changing the mathematical interpolant. Constant-

Figure 3a: Standard tomography
Partial reproduction: independently generated geometry and explicit numerical substitutions

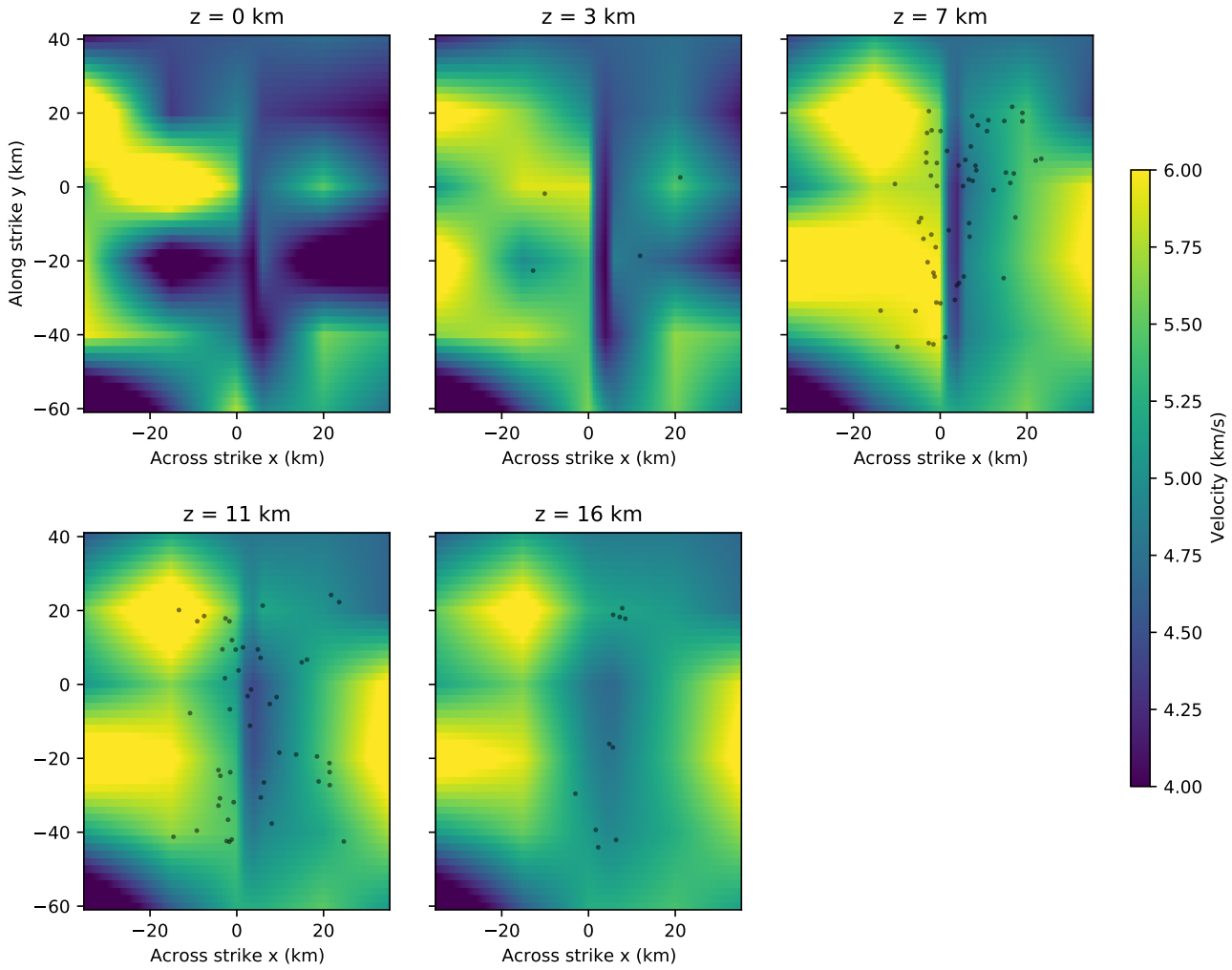


Figure 4: Partial reconstruction of manuscript Figure 3a: standard tomographic velocity from independent observations. Dots are inferred sources selected by adjacent depth midpoints. Colors show 4–6 km/s; values outside this interval saturate and their numerical range is reported in the text. Unconstrained boundaries must not be read as resolved structure.

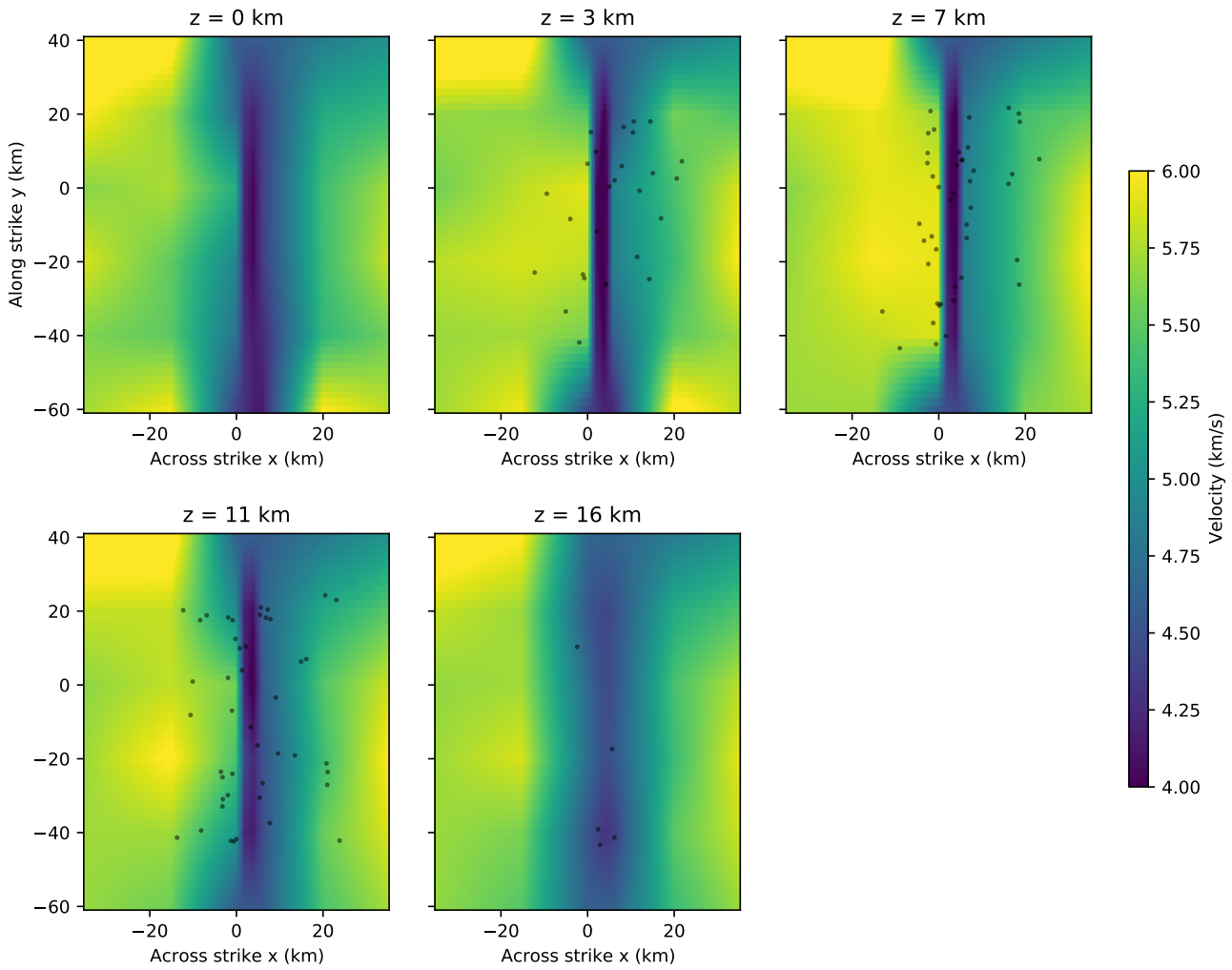


Figure 5: Partial reconstruction of manuscript Figure 3b: DD velocity from independent observations. The initial model, true medium, noise, and inversion grid are identical to those of the standard run. Colors and source-selection conventions follow the preceding figure.

Figure 3c: True minus DD velocity
Partial reproduction: independently generated geometry and explicit numerical substitutions

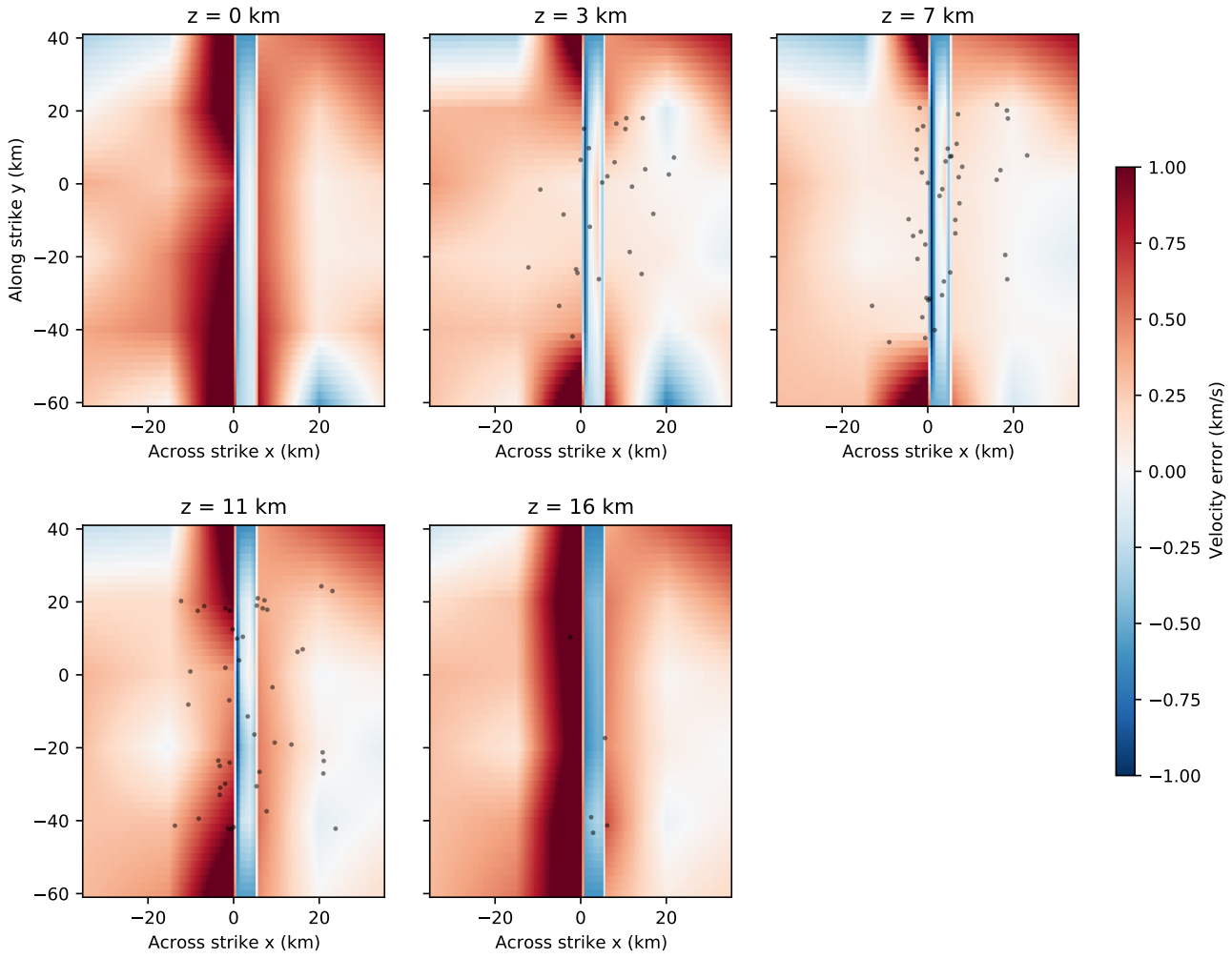


Figure 6: Partial reconstruction of manuscript Figure 3c: true minus DD velocity, following the subtraction order in the manuscript's prose. Blue marks recovered velocities that are too fast, and red marks recovered velocities that are too slow. The fixed color range is ±1 km/s. Saved samples retain unsaturated values.

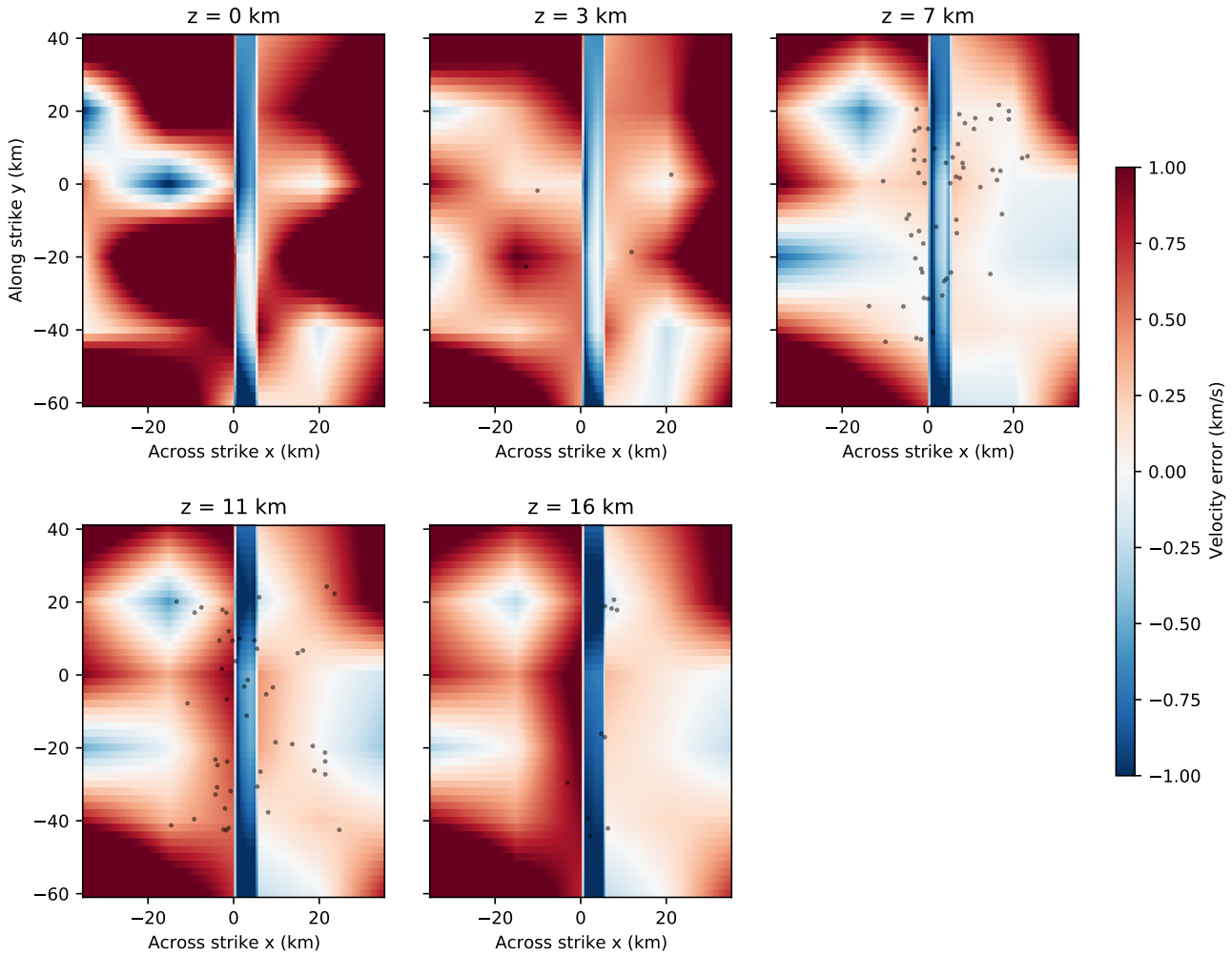


Figure 7: Partial reconstruction of manuscript Figure 3d: true minus standard tomographic velocity, with the sign and color convention of the preceding figure. The spatial patterns depend on the independent geometry and cannot be compared point by point with the original inversion.

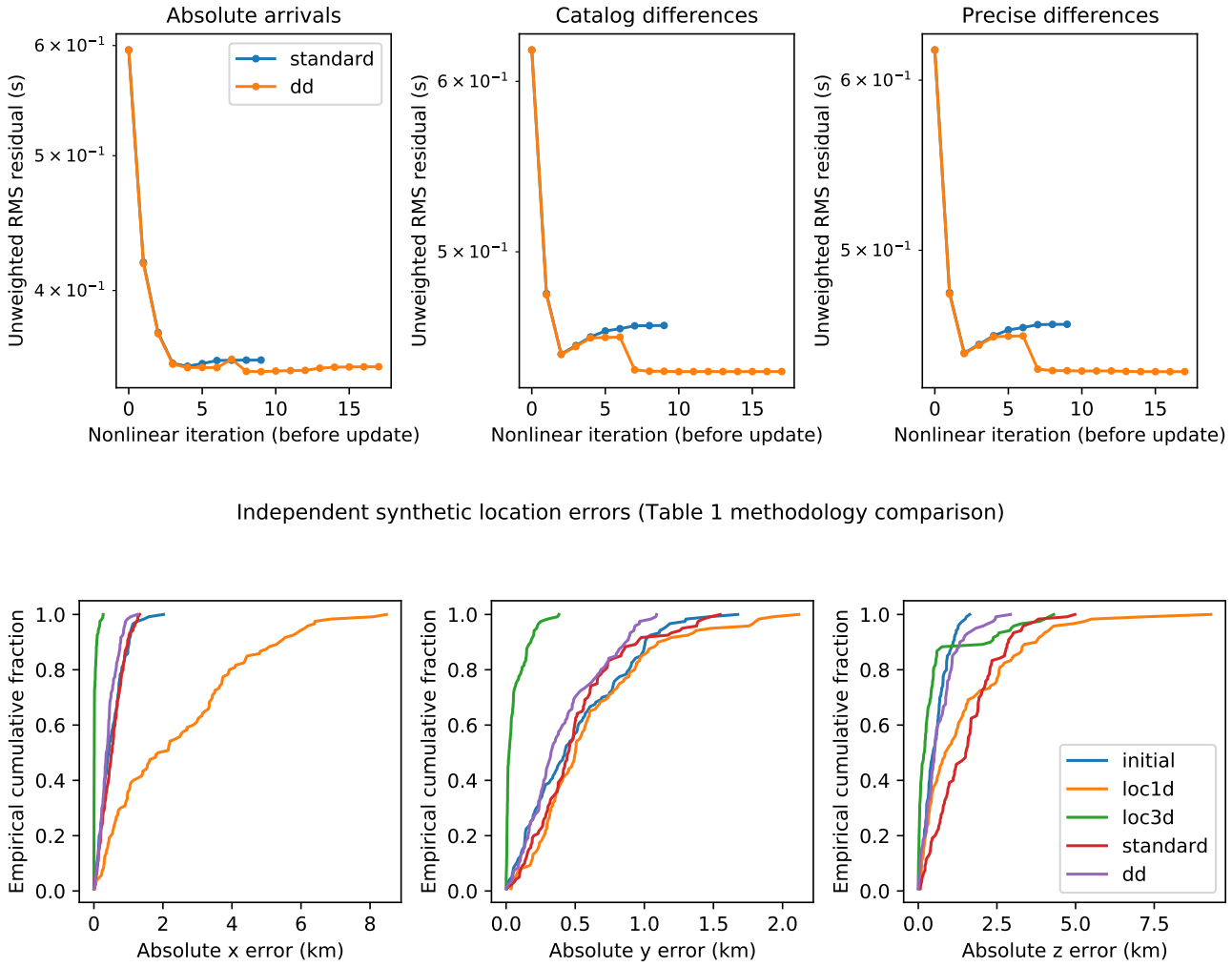


Figure 8: Independent convergence and location-error diagnostics. Top: RMS residuals before each nonlinear update. Stage transitions change the objective, so the unweighted RMS can rise while a weighted objective falls. Bottom: source-coordinate error distributions against saved true coordinates. These are additional method tests, not field-data panels.

axis and surface-ray regression checks cover the correction. The earlier numerical histories are retained in `data/development/`; they are not used for any final numerical panel. Domain projection does not make sources pinned to boundaries physically reliable. In the depth-independent true medium, a source at the surface has zero first-order depth sensitivity to the surface receivers. A linearized fixed-model relocation therefore cannot move such a source back into the interior. The boundary counts record this remaining geometric limitation; the symmetry correction removes numerical drift rather than supplying missing depth sensitivity.

For Hayward Figures 4–11, the missing dependencies are observational. The field velocity model is the result to be inferred, not an input that can be reconstructed by assuming the reported contrast. We recovered the written grid, projection conventions, sample counts and section definitions, then traced each panel's required data. Those initial attempts stopped at excluded arrivals and event coordinates. In particular, resolution and uncertainty require the actual linearized ray matrix, damping convention and covariance model. A synthetic resolution matrix would not reproduce Figure 11, so none is substituted here.

6 Rebuild and artifact provenance

Run the following command from the project root:

```
./rebuild.sh
```

It configures and builds C++, runs validation, generates observations, performs four main inversion comparisons, two starting-profile sensitivities, fixed-model and joint recovery tests, and true/final-model ray checks, computes metrics, generates all figures and audit tables, and compiles this report twice with pdfLaTeX. Its outcome is recorded in `data/rebuild_result.txt` only when all commands have succeeded. Machine-readable numerical checks and iteration histories remain in `data/`; raw console logs are local diagnostics excluded from the shared repository. All caches and build intermediates are confined to the project directory.

For the baseline delivery, the complete clean procedure finished successfully on 9 October 2026 in 801 seconds, with all 32 implementation consistency checks passing and both PDF compilation passes completing. The numerical results come from that run. `reference/delivery_environment.json` records the compiler, Python packages, TeX version, and parallel settings. Interpretation and plot-label revisions were compiled afterward from those same numerical outputs. The subsequent ray-kernel changes described below have not been followed by a complete inversion rebuild. The 801-second record therefore verifies the baseline, not the revised kernel.

Requirements are a C++17 compiler, CMake, OpenMP, Python with NumPy and Matplotlib, and pdfLaTeX with standard packages. No tomography, optimization, or authors' software dependency is required. The saved input CSV files permit rerunning the inverse problem without regenerating random numbers. Standard-library random distributions may differ across platforms, so cross-platform bitwise equality of newly generated observations is not assumed. The experiment configuration, source code, plotting scripts, generated data and figure audit are retained with this LaTeX source.

7 Resumed work: authoritative inputs and ray corrections

On 9 October 2026 the user required real earthquake and station observations and metadata from authoritative sources, beginning with USGS and EarthScope. Saved records must retain SEED naming codes and resource identifiers. USGS supplied a Loma Prieta mainshock QuakeML product containing 315 P picks; EarthScope supplied 364 historical NC station records. Provider identifiers, station epochs, retrieval URLs, raw-file hashes, and identifier attachments are saved in `data/authoritative/discovery/`. These are archive-access checks, not the manuscript's selected event and station set.

The target's exact Loma Prieta selection remains unidentified. Its Hayward sample contains 1,489 events and 52 stations, whereas the cited precursor describes a broader selection. The user was asked whether to trace the exact selection further, supply identifiers, or permit a documented independent subset. Dataset selection and inversion remain stopped under the explicit clarification rule. The field-panel statuses therefore have not changed.

Independent component work isolated two numerical deficiencies. First, two-point integration of a segment crossing velocity-grid boundaries left a 1.19×10^{-4} s/km discrepancy between its variational endpoint gradient and finite differences of computed time. Eight-point integration reduces that example's discrepancy to 5.6×10^{-12} s/km and its time error from 33.4 microseconds to numerical roundoff. A separate three-dimensional segment check gives 8.9×10^{-11} s/km. This improves integration accuracy; it is not proof of exact gradients for every finite quadrature problem.

Second, the homogeneous transverse ray Hessian is a fixed-endpoint path Laplacian, whose condition number grows quadratically with path-node count. Replacing the scalar initial L-BFGS metric with an inverse weighted path Laplacian reduces this stiffness without changing the Fermat objective or admissible paths. Positive segment weights are midpoint slowness divided by segment length. An analytic circular-ray test in $v(z) = 5 + 0.2z$ km/s over 40 km failed to converge within 1,000 baseline updates at spacings 1, 0.5, and 0.25 km. The revised metric converges in 32, 23, and 25 updates with unchanged stopping criteria. Halving spacing reduces the travel-time error by approximately four. A metric-only control preserves this improvement, separating it from the quadrature correction.

All 48 current component checks pass. Forward retracing of frozen baseline models gives the following convergence counts out of 32 sampled rays:

Frozen model	Production: baseline / revised	Refined: baseline / revised
Standard	1 / 5	3 / 22
Double difference	1 / 9	8 / 23

Nonconvergence remains substantial, and some source finite-difference errors increase despite improved stationarity. These checks do not demonstrate that the 0.439-s baseline DD differential residual is corrected. No inverse solution or manuscript panel was regenerated with the revised kernel. Analytic comparisons, source hashes, all 48 checks, and frozen-model ray tables are in `data/development/ray_optimizer_audit/`. Run the standalone audit with

```
python3 diagnostics/run_ray_optimizer_audit.py
```

Its README records the validation and forward-check commands. The appendix below retains the baseline 32-check record supporting the baseline figures.

A Numerical validation record

Check	Error	Tolerance
homogeneous_travel_time	4.44e-15	1e-10
homogeneous_source_derivative	5.19e-17	1e-09
homogeneous_velocity_derivative	4.44e-16	1e-09
partition_of_unity	1.42e-14	1e-09
ray_reciprocity	6.58e-07	2e-05
bending_not_slower_than_straight	0	1e-10
ray_refinement_seconds	4.14e-05	0.015
bent_source_finite_difference	9.22e-05	0.003
velocity_finite_difference	3.71e-13	1e-07
identical_event_dd_cancellation	0	1e-12
station_bias_cancellation	0	1e-12
lsqr_known_solution	7.11e-16	1e-09
matrix_adjoint	0	1e-12
interpolation_constant_x_symmetry	0	0
interpolation_constant_y_symmetry	0	0
interpolation_constant_z_symmetry	0	0
surface_ray_plane_symmetry	0	0
surface_ray_source_symmetry	0	0
snell_reference_travel_time	6.9e-05	0.001
snell_reference_source_derivative	8e-06	0.0001
trilinear_gradient_finite_difference	3.52e-12	2e-10
boundary_ray_feasible_descent	0	1e-10
boundary_ray_projected_gradient	1.84e-06	1e-05
boundary_ray_domain	0	0
collapsed_vertices_preserve_endpoints	0	0
collapsed_ray_minimum_segment	0	0
collapsed_ray_gradient_regression	4.53e-05	0.001
scaled_damped_lsqr_solution	2.48e-16	1e-10
scaled_damped_lsqr_normal_residual	5.55e-17	1e-08
scaled_damped_lsqr_converged	0	0
zero_rhs_lsqr	0	0
truncated_lsqr_not_converged	0	0

Error units follow each test: seconds for time comparisons, seconds per km for source derivatives, seconds per (km/s) for velocity derivatives, and algebraic units for the sparse-system checks. These tolerances were chosen for numerical consistency tests, not to enforce agreement with the paper.

B Figure-by-figure record

The complete machine-readable record is in `data/panel_audit.csv` and `data/panel_audit.json`; its readable counterpart is `FIGURE_AUDIT.md`. "Independently reproduced" means the quantity is reconstructed from the written specification using our model output. "Partially reproduced" means an independent numerical counterpart exists but one or more specified experimental inputs or algorithms differ. "Not reproduced" means no corresponding result is claimed.

Panel	Quantity	Status and limiting dependency
1	Horizontal true P-velocity slice	independently reproduced. Physical velocity profile reconstructed from prose; plotting extent and depth chosen from model domain.
2	Synthetic-test event/station geometry and inversion grid	partially reproduced. Grid recovered from caption. Original Loma Prieta coordinates are excluded and are not reconstructed from map pixels.
3a	Standard tomographic P velocity; all generated depth slices	partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
3b	DD tomographic P velocity; all generated depth slices	partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
3c	True minus DD P velocity; all generated depth slices	partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
3d	True minus standard P velocity; all generated depth slices	partially reproduced. Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
4	Hayward network, catalog seismicity, and inversion grid	not reproduced. Attempt: recovered written grid and required observation counts; no authorized coordinates for 1489 events or 52 stations. No invented field map is presented.
5a	Hayward event map: NCSN catalog	not reproduced. Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.
5b	Hayward event map: standard tomography	not reproduced. Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.
5c	Hayward event map: DD location	not reproduced. Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.

Panel	Quantity	Status and limiting dependency
5d	Hayward event map: DD tomography	not reproduced. Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.
6a	catalog: geographical map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
6b	catalog: rotated Cartesian map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
6c	catalog: across-strike depth section AA-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
6d	catalog: along-strike depth section BB-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
7a	standard tomography: geographical map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
7b	standard tomography: rotated Cartesian map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
7c	standard tomography: across-strike depth section AA-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
7d	standard tomography: along-strike depth section BB-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.

Panel	Quantity	Status and limiting dependency
8a	DD location: geographical map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
8b	DD location: rotated Cartesian map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
8c	DD location: across-strike depth section AA-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
8d	DD location: along-strike depth section BB-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
9a	DD tomography: geographical map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
9b	DD tomography: rotated Cartesian map	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
9c	DD tomography: across-strike depth section AA-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
9d	DD tomography: along-strike depth section BB-prime	not reproduced. Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
10a (y=-30)	Hayward across-strike DD velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.

Panel	Quantity	Status and limiting dependency
10a (y=-20)	Hayward across-strike DD velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10a (y=-10)	Hayward across-strike DD velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10a (y=0)	Hayward across-strike DD velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10a (y=10)	Hayward across-strike DD velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10a (y=20)	Hayward across-strike DD velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10a (y=30)	Hayward across-strike DD velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10b (y=-30)	Hayward across-strike standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10b (y=-20)	Hayward across-strike standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10b (y=-10)	Hayward across-strike standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.

Panel	Quantity	Status and limiting dependency
10b (y=0)	Hayward across-strike standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10b (y=10)	Hayward across-strike standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10b (y=20)	Hayward across-strike standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10b (y=30)	Hayward across-strike standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10c (y=-30)	Hayward across-strike DD minus standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10c (y=-20)	Hayward across-strike DD minus standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10c (y=-10)	Hayward across-strike DD minus standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10c (y=0)	Hayward across-strike DD minus standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10c (y=10)	Hayward across-strike DD minus standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.

Panel	Quantity	Status and limiting dependency
10c (y=20)	Hayward across-strike DD minus standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
10c (y=30)	Hayward across-strike DD minus standard velocity	not reproduced. Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
11a (y=-30)	Hayward resolution-matrix diagonal	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11a (y=-20)	Hayward resolution-matrix diagonal	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11a (y=-10)	Hayward resolution-matrix diagonal	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11a (y=0)	Hayward resolution-matrix diagonal	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11a (y=10)	Hayward resolution-matrix diagonal	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11a (y=20)	Hayward resolution-matrix diagonal	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11a (y=30)	Hayward resolution-matrix diagonal	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.

Panel	Quantity	Status and limiting dependency
11b (y=-30)	Hayward one-sigma velocity uncertainty	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11b (y=-20)	Hayward one-sigma velocity uncertainty	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11b (y=-10)	Hayward one-sigma velocity uncertainty	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11b (y=0)	Hayward one-sigma velocity uncertainty	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11b (y=10)	Hayward one-sigma velocity uncertainty	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11b (y=20)	Hayward one-sigma velocity uncertainty	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
11b (y=30)	Hayward one-sigma velocity uncertainty	not reproduced. Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.

B.1 Specifications, commands, and attempted dependencies

The following records specify the source description, parameters, generation command, and plot command for each panel. Seven sections of a common Figure 10 or 11 panel share the explicitly listed settings. Unavailable commands are reported as unavailable rather than replaced by placeholders that would fabricate field observations.

Panel 1

Quantity: Horizontal true P-velocity slice
Written specification: Synthetic Test, p.1877; caption 1
Parameters: Written sandwich profile, x=-35..35 km, y=-60..40 km; constant with depth

Status: independently reproduced
Attempt and limitation: Physical velocity profile reconstructed from prose; plotting extent and depth chosen from model domain.
Generate data:

```
./build/dd generate configs/synthetic.cfg data/synthetic
```

Create plot:

```
python3 plot.py --figure 1
```

Panel 2

Quantity: Synthetic-test event/station geometry and inversion grid
Written specification: Caption 2, p.1879
Parameters: Published 8x6x5 inversion grid; 120 independent sources and 32 independent stations, seed 2003
Status: partially reproduced
Attempt and limitation: Grid recovered from caption. Original Loma Prieta coordinates are excluded and are not reconstructed from map pixels.
Generate data:

```
./build/dd generate configs/synthetic.cfg data/synthetic
```

Create plot:

```
python3 plot.py --figure 2
```

Panel 3a

Quantity: Standard tomographic P velocity; all generated depth slices
Written specification: Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
Parameters: Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
Status: partially reproduced
Attempt and limitation: Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
Generate data:

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    standard
```

Create plot:

```
python3 plot.py --figure 3a
```

Panel 3b

Quantity: DD tomographic P velocity; all generated depth slices
Written specification: Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
Parameters: Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
Status: partially reproduced
Attempt and limitation: Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
Generate data:

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    dd
```

Create plot:

```
python3 plot.py --figure 3b
```

Panel 3c

Quantity: True minus DD P velocity; all generated depth slices
Written specification: Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
Parameters: Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
Status: partially reproduced
Attempt and limitation: Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
Generate data:

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    dd
```

Create plot:

```
python3 plot.py --figure 3c
```

Panel 3d

Quantity: True minus standard P velocity; all generated depth slices
Written specification: Synthetic Test pp.1877-1879; captions 3 pp.1880-1881
Parameters: Depths 0,3,7,11,16 km; grid from caption 2; noise 0.04 s and station bias +/-0.3 s; configs/synthetic.cfg
Status: partially reproduced
Attempt and limitation: Independent geometry, homogeneous initial velocity, and explicit replacements for unpublished numerical settings; numerical comparisons cannot isolate implementation from experiment differences.
Generate data:

```
./build/dd generate configs/synthetic.cfg data/synthetic && ./build/dd invert configs/synthetic.cfg data/synthetic
    standard
```

Create plot:

```
python3 plot.py --figure 3d
```

Panel 4

Quantity: Hayward network, catalog seismicity, and inversion grid
Written specification: Caption 4, p.1882
Parameters: Grid x=-35,-20,-10,-5,-4,-3,0,5,10,20,35; y=-30,-20,-10,0,10,20,30; z=0,4,8,12,20 km
Status: not reproduced
Attempt and limitation: Attempt: recovered written grid and required observation counts; no authorized coordinates for 1489 events or 52 stations. No invented field map is presented.
Generate data:

```
Unavailable: requires excluded observed coordinates
```

Create plot:

```
Unavailable: no source data
```

Panel 5a

Quantity: Hayward event map: NCSN catalog
Written specification: Event Locations pp.1881-1883; caption 5
Parameters: 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
Status: not reproduced
Attempt and limitation: Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.
Generate data:

```
Unavailable: requires excluded observations
```

Create plot:

```
Unavailable: no source data
```

Panel 5b

Quantity: Hayward event map: standard tomography
Written specification: Event Locations pp.1881-1883; caption 5
Parameters: 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
Status: not reproduced
Attempt and limitation: Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.
Generate data:

```
Unavailable: requires excluded observations
```

Create plot:

```
Unavailable: no source data
```

Panel 5c

Quantity: Hayward event map: DD location
Written specification: Event Locations pp.1881-1883; caption 5
Parameters: 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
Status: not reproduced
Attempt and limitation: Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.
Generate data:

```
Unavailable: requires excluded observations
```

Create plot:

```
Unavailable: no source data
```

Panel 5d

Quantity: Hayward event map: DD tomography
Written specification: Event Locations pp.1881-1883; caption 5
Parameters: 1489 earthquakes, 52 stations; 20257 absolute picks, 767127 catalog differences, 17955 precise differences
Status: not reproduced
Attempt and limitation: Attempt: traced input dependencies. Catalog picks, differential times and event identities cannot be inferred from written descriptions or regenerated by a physical forward model.
Generate data:

```
Unavailable: requires excluded observations
```

Create plot:

```
Unavailable: no source data
```

Panel 6a

Quantity: catalog: geographical map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 6b

Quantity: catalog: rotated Cartesian map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 6c

Quantity: catalog: across-strike depth section AA-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 6d

Quantity: catalog: along-strike depth section BB-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 7a

Quantity: standard tomography: geographical map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 7b

Quantity: standard tomography: rotated Cartesian map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 7c

Quantity: standard tomography: across-strike depth section AA-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 7d

Quantity: standard tomography: along-strike depth section BB-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 8a

Quantity: DD location: geographical map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 8b

Quantity: DD location: rotated Cartesian map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 8c

Quantity: DD location: across-strike depth section AA-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 8d

Quantity: DD location: along-strike depth section BB-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 9a

Quantity: DD tomography: geographical map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 9b

Quantity: DD tomography: rotated Cartesian map
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 9c

Quantity: DD tomography: across-strike depth section AA-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 9d

Quantity: DD tomography: along-strike depth section BB-prime
Written specification: Event Locations pp.1882-1884; captions 6-9
Parameters: Center 37.72 N, 122.06 W; 35-degree anticlockwise rotation; map selection 37.83..37.91 N, -122.26..-122.18 E
Status: not reproduced
Attempt and limitation: Attempt: recovered projection convention and regional selection. Input event locations/identities and, for c/d, box membership are unavailable under the data restriction. No digitization used.
Generate data:

```
Unavailable: requires excluded event catalog and travel times
```

Create plot:

```
Unavailable: no source data
```

Panel 10a: all seven sections at y=-30,-20,-10,0,10,20,30 km

Quantity: Hayward across-strike DD velocity
Written specification: Velocity Structure p.1884; caption 10 p.1887
Parameters: Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
Status: not reproduced
Attempt and limitation: Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
Generate data:

```
Unavailable: requires excluded Hayward observations
```

Create plot:

```
Unavailable: no field inversion output
```

Panel 10b: all seven sections at y=-30,-20,-10,0,10,20,30 km

Quantity: Hayward across-strike standard velocity
Written specification: Velocity Structure p.1884; caption 10 p.1887
Parameters: Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
Status: not reproduced
Attempt and limitation: Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
Generate data:

```
Unavailable: requires excluded Hayward observations
```

Create plot:

```
Unavailable: no field inversion output
```

Panel 10c: all seven sections at y=-30,-20,-10,0,10,20,30 km

Quantity: Hayward across-strike DD minus standard velocity
Written specification: Velocity Structure p.1884; caption 10 p.1887
Parameters: Published Hayward grid; smoothing weight 10; all seven y sections listed above; hypocenters within 5 km
Status: not reproduced
Attempt and limitation: Attempt: recovered grid, smoothing choice and cross-section convention. The observational velocity contrast is an inversion result, not a forward-model input specified by prose.
Generate data:

```
Unavailable: requires excluded Hayward observations
```

Create plot:

```
Unavailable: no field inversion output
```

Panel 11a: all seven sections at y=-30,-20,-10,0,10,20,30 km

Quantity: Hayward resolution-matrix diagonal
Written specification: Velocity Structure pp.1884,1886; caption 11 p.1888
Parameters: Absolute-data standard-tomography linearization at the DD velocity model; all seven y sections listed above
Status: not reproduced
Attempt and limitation: Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
Generate data:

```
Unavailable: requires excluded observation matrix and inferred field velocity
```

Create plot:

```
Unavailable: no field matrix
```

Panel 11b: all seven sections at y=-30,-20,-10,0,10,20,30 km

Quantity: Hayward one-sigma velocity uncertainty
Written specification: Velocity Structure pp.1884,1886; caption 11 p.1888
Parameters: Absolute-data standard-tomography linearization at the DD velocity model; all seven y sections listed above
Status: not reproduced
Attempt and limitation: Attempt: established dependency on the actual ray geometry, final field model, damping and data covariance. These cannot be replaced by an unrelated synthetic resolution matrix and called a reproduction.
Generate data:

```
Unavailable: requires excluded observation matrix and inferred field velocity
```

Create plot:

```
Unavailable: no field matrix
```

C References and guidance

Zhang, H., and C. H. Thurber (2003). Double-Difference Tomography: The Method and Its Application to the Hayward Fault, California. *BSSA*, 93, 1875–1889. doi:10.1785/0120020190.

Oppenheimer, D. H., F. W. Klein, J. P. Eaton, and F. W. Lester (1993). The Northern California Seismic Network bulletin, January–December 1992. USGS Open-File Report 93-578, Table 4. doi:10.3133/ofr93578.

The two user-permitted guidance files are repository guidelines and academic style guide. They govern research-code simplicity and prose. They were not used as permission to inspect any original model implementation or output.